

Basic Information

Experiment No.: 07

Date: 24-09-2026

Title: Java Vector and Its Methods

Objective / Aim

To implement the "Vector" class in Java and demonstrate adding, accessing, updating, removing, searching, size, and capacity operations.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

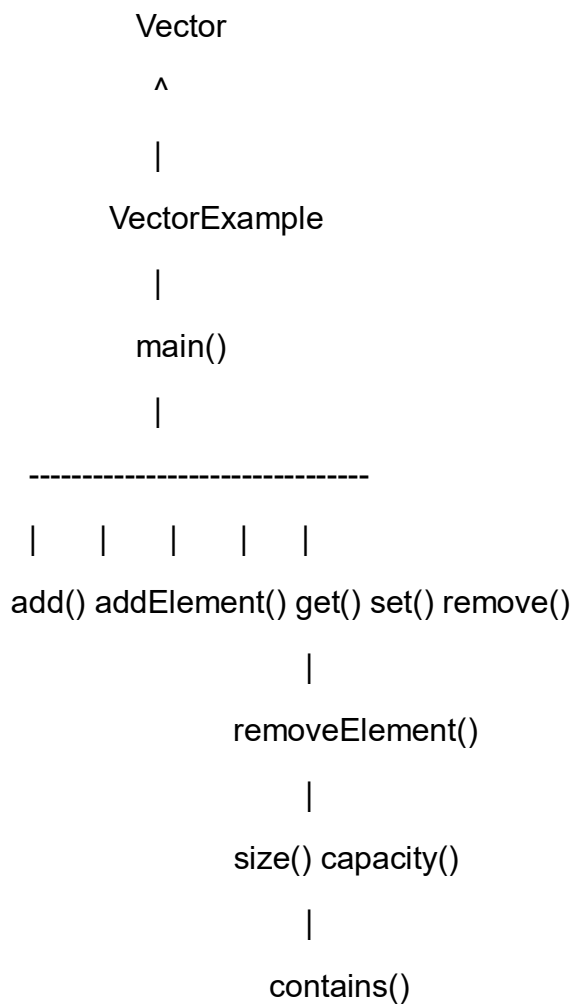
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "Vector" of strings.
2. Add elements using "add()".
3. Add an element using "addElement()".
4. Access an element using "get()".
5. Update an element using "set()".

6. Remove an element using "remove()".
 7. Remove a specific element using "removeElement()".
 8. Find the number of elements using "size()".
 9. Display the vector capacity using "capacity()".
 10. Check an element using "contains()".
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UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.Vector;
```

```
public class VectorExample {
```

```
    public static void main(String[] args) {
```

```
        Vector<String> vector = new Vector<>();
```

```
        // add()
```

```
        vector.add("Apple");
```

```
        vector.add("Banana");
```

```
        vector.add("Mango");
```

```
        System.out.println("Vector: " + vector);
```

```
        // addElement()
```

```
        vector.addElement("Orange");
```

```
        System.out.println("After addElement: " + vector);
```

```
        // get()
```

```
        System.out.println("Element at index 1: " +  
            vector.get(1));
```

```
        // set()
```

```
        vector.set(1, "Grapes");
```

```
        System.out.println("After set: " + vector);
```

```
// remove(index)
vector.remove(2);
System.out.println("After remove: " + vector);

// removeElement()
vector.removeElement("Apple");
System.out.println("After removeElement: " + vector);

// size()
System.out.println("Size: " + vector.size());

// capacity()
System.out.println("Capacity: " + vector.capacity());

// contains()
System.out.println("Contains Grapes: " +
    vector.contains("Grapes"));
}
}
```

Compilation & Execution Commands

Save the file as:

VectorExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac VectorExample.java
```

Run:

```
java collectionframework.VectorExample
```

Output Screen

Vector: [Apple, Banana, Mango]

After addElement: [Apple, Banana, Mango, Orange]

Element at index 1: Banana

After set: [Apple, Grapes, Mango, Orange]

After remove: [Apple, Grapes, Orange]

After removeElement: [Grapes, Orange]

Size: 2

Capacity: 10

Contains Grapes: true

Result

The "Vector" class was successfully implemented, and operations such as "add()", "addElement()", "get()", "set()", "remove()", "removeElement()", "size()", "capacity()", and "contains()" were successfully demonstrated.